

PEAK ACHIEVERS
ONLINE ACADEMY

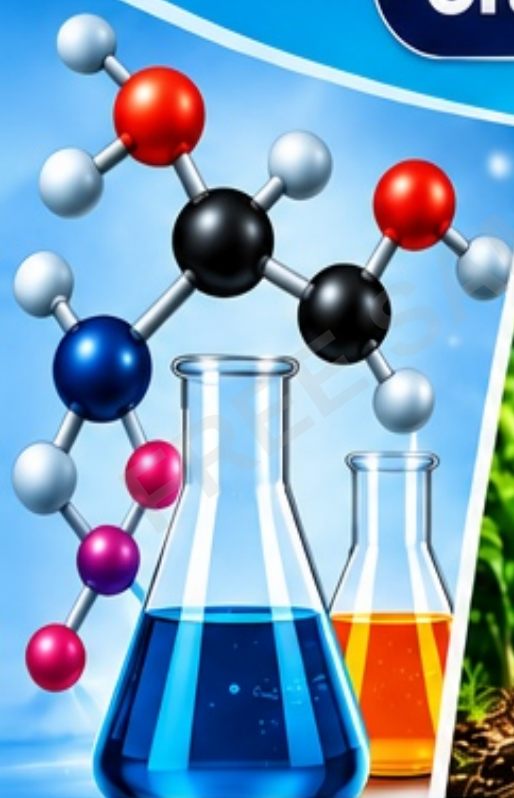
INTEGRATED SCIENCE

**REVISION NOTES • TOPIC QUIZZES • MIXED CONCEPTS
CBE MODEL TESTS • COMPLETE MARKING SCHEMES**

Grade

8

BOOK 2



**Mixtures, Elements
and Compounds**



**Living Things and
their Environment**



Force and Energy

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Dedication

In loving memory of my late father, Simeon Nyatuka, whose guidance, sacrifice and inspiration remain part of my educational journey.

With heartfelt appreciation to Hyrine Kerubo, Peter Munde Mong'are, Teresa Kwamboka Sawe, Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani, and all those whose support and encouragement have strengthened this journey.

To Her, my special one, for being a special part of my journey and for the place you hold in my heart.



To my fellow teachers, guardians, family and every learner who continues to inspire the work behind Peak Achievers.

May every learner using this book remain curious, work with determination, and discover the confidence to learn, apply, think big and achieve.

How to Use This Book

REVISE → PRACTISE → APPLY → THINK BIG → ASSESS → CHECK → ACHIEVE

Study the brief Q&A notes, complete each topical quiz, then attempt the strand quizzes, Mixed Concepts Quizzes and final model tests. Show working in calculations and use correct scientific terms and units.

Grade 8 Scope

Strand 1: Mixtures, Elements and Compounds

1.1 Elements and Compounds • 1.2 Physical and Chemical Changes • 1.3 Classes of Fire

Strand 2: Living Things and their Environment

2.1 The Cell • 2.2 Movement of Materials In and Out of the Cell • 2.3 Reproduction in Human Beings

Strand 3: Force and Energy

3.1 Transformation of Energy • 3.2 Pressure

STRAND 1: MIXTURES, ELEMENTS AND COMPOUNDS

1.1 Elements and Compounds

Q. What is matter?

A. Anything that has mass and occupies space.

Q. What is an atom?

A. The smallest particle of an element that takes part in a chemical reaction.

Q. What is an element?

A. A pure substance made of only one type of atom.

Q. What is a molecule?

A. Two or more atoms chemically bonded together.

Q. What is a compound?

A. A pure substance containing two or more different elements chemically combined in fixed proportions.

Q. What is a mixture?

A. Two or more substances physically combined; the components are not chemically bonded.

Q. Why is O_2 a molecule but not a compound?

A. It has two bonded atoms, but both are oxygen atoms.

Q. What are reactants and products?

A. Reactants are starting substances; products are substances formed.

Q. Give common symbols.

A. H, O, C, N, Na, Cl, Al, Fe, Cu, Ag, Au, Hg and Pb.

Q. What is the symbol rule?

A. First letter capital; second letter, if present, lower-case.

Common Chemical Symbols

H Hydrogen • O Oxygen • C Carbon • N Nitrogen • Na Sodium • Cl Chlorine • Al Aluminium • Fe Iron • Cu Copper • Ag Silver • Au Gold • Hg Mercury • Pb Lead

End-of-Topic Quiz

1. Define element. (2)

2. Compound vs mixture. (3)
3. Write Na, Fe, Cu, Cl. (4)
4. Why O₂ molecule not compound? (3)
5. Classify water, air, copper. (3)
6. Write carbon + oxygen → carbon dioxide. (2)
7. State symbol capitalisation rule. (2)
8. Explain sand+salt as mixture. (3)
9. Define reactant/product. (4)

Answer Guide

Use the Q&A notes immediately above. For calculations: $P=F/A$; $F=PA$; $A=F/P$. Show formula, substitution, answer and unit. For practical questions, identify variables, describe the trend and give a scientific conclusion.

1.2 Physical and Chemical Changes

Q. What is a physical change?

A. A change in form, size or state without forming a new substance.

Q. What is a chemical change?

A. A change that forms one or more new substances.

Q. Name common state changes.

A. Melting, freezing, evaporation/boiling and condensation.

Q. How are particles in a solid arranged?

A. Closely packed and vibrating about fixed positions.

Q. How do liquid particles behave?

A. They are close together but move past one another.

Q. How do gas particles behave?

A. They are far apart and move rapidly in many directions.

Q. What does heating generally do?

A. It increases particle kinetic energy.

Q. What can a heating plateau indicate?

A. A change of state; supplied energy is used in the state change rather than raising temperature.

Q. What can indicate chemical change?

A. Gas formation, colour change, new solid, heat/light change or a new smell; the key criterion is formation of new substance(s).

End-of-Topic Quiz

1. Define physical and chemical changes. (4)

2. Name four changes of state. (4)

3. Why melting ice is physical. (2)

4. Three indicators chemical change. (3)

5. Describe particles in solid/liquid/gas. (6)

6. Explain heating plateau. (3)

7. Classify rusting, boiling, burning. (3)

8. Heating effect on particles. (2)

Answer Guide

Use the Q&A notes immediately above. For calculations: $P=F/A$; $F=PA$; $A=F/P$. Show formula, substitution, answer and unit. For practical questions, identify variables, describe the trend and give a scientific conclusion.

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1.3 Classes of Fire

Q. What forms the fire triangle?

A. Heat, fuel and oxygen.

Q. What is cooling?

A. Reducing heat.

Q. What is smothering?

A. Reducing oxygen.

Q. What is starvation?

A. Removing or isolating fuel.

Q. What is Class A?

A. Ordinary solid combustibles.

Q. What is Class B?

A. Flammable liquids.

Q. What is Class C in this book?

A. Flammable gases.

Q. What is Class D?

A. Combustible metals.

Q. What is Class F?

A. Cooking oils and fats.

Q. Why not water on cooking oil?

A. It can make hot burning oil splash or spread violently.

Q. What is the emergency sequence?

A. Raise alarm → evacuate safely → assembly point → follow instructions.

Q. What if clothing catches fire?

A. STOP → DROP → ROLL.

CAUTION: Fire-class labels and extinguisher systems can vary. Follow school and relevant local safety guidance.

End-of-Topic Quiz

1. Fire triangle. (3)
2. Cooling/smothering/starvation. (6)
3. Class A and F. (2)
4. Why no water on cooking oil? (2)
5. Four emergency actions. (4)
6. Stop-Drop-Roll. (3)
7. Why exits clear? (2)
8. Why fuel type matters? (3)
9. One classification caution. (2)

Answer Guide

Use the Q&A notes immediately above. For calculations: $P=F/A$; $F=PA$; $A=F/P$. Show formula, substitution, answer and unit. For practical questions, identify variables, describe the trend and give a scientific conclusion.

2.3 Reproduction in Human Beings

Q. What is puberty?

A. A period of physical and emotional changes associated with sexual maturity.

Q. Function of testes?

A. Produce sperm and male sex hormones.

Q. Function of ovaries?

A. Produce ova and female sex hormones.

Q. What is ovulation?

A. Release of a mature ovum from an ovary.

Q. Where does fertilisation normally occur?

A. In the oviduct/Fallopian tube.

Q. What is fertilisation?

A. Fusion of male and female gametes to form a zygote.

Q. Give the sequence after fertilisation.

A. Fertilisation → zygote → embryo → implantation → foetus.

Q. Where does implantation occur?

A. In the uterine lining.

Q. Placenta function?

A. Exchange of substances between mother and foetus while circulations remain distinct.

Q. Umbilical cord function?

A. Connects the foetus to the placenta and carries blood in its vessels.

Q. Amniotic fluid function?

A. Cushions and protects the foetus.

Q. Is every cycle exactly 28 days?

A. No. Menstrual-cycle length varies.

Q. What is HIV?

A. A virus that attacks the immune system.

Q. What is AIDS?

A. An advanced stage of HIV infection associated with severe immune-system weakening.

Q. Can HIV spread by ordinary social contact?

A. No.

End-of-Topic Quiz

1. Define puberty. (2)
2. Testes/ovaries functions. (4)
3. Define ovulation. (2)
4. Site fertilisation. (1)
5. Sequence to foetus. (5)
6. Placenta/cord/amniotic fluid. (6)
7. Cycle variation. (2)
8. HIV vs AIDS. (3)
9. Social-contact myth. (2)

Answer Guide

Use the Q&A notes immediately above. For calculations: $P=F/A$; $F=PA$; $A=F/P$. Show formula, substitution, answer and unit. For practical questions, identify variables, describe the trend and give a scientific conclusion.

STRAND 3 QUIZ — 50 MARKS

1. Energy forms/transformations. (12)
2. Conservation and renewable/non-renewable sources. (10)
3. Pressure definition/formula/calculations. (12)
4. Liquid-pressure practical. (6)
5. Solar-energy and rescue-vehicle CBE scenario. (10)

Answer Guide

Award marks for correct scientific terms, reasoning and calculations. Refer to the preceding strand notes and topical answer guides.

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